Flood Alerts Work. People Still Die.

An internal briefing on the gap between warning and action – and what we can do about it.

The Alert System Is Solved

What Already Exists

India's flood warning infrastructure is more capable than most people realize. The warning *generation* problem is largely solved:

ASDMA / FLEWS

Generates river-level forecasts for all major waterways in Assam

Sachet App (NDMA)

Pushes alerts nationally across **12 languages** to registered users

Cell Broadcast System

Sends loud pop-up alerts directly to phones – no app required

Upper Assam, July 2024: Where It Broke

Coverage Gap

Cell Broadcast was only a **Kamrup Metro pilot** – it never reached the hardest-hit districts in Upper Assam

Language Barrier

Alerts that did go out were **English-only at first** – a real barrier for Assamese and local-language speakers in rural communities

The Unanswered Question

People got warned. **Then what?** No clear guidance on where to go, which roads were open, or where shelter was available

The Real Gap: Alert to Action

Warning sent ≠ People safe

Generating and sending a warning is solved. **Everything after the alert fires is broken.**

Getting people to safety requires answers to questions the alert doesn't address:

- Which roads are still passable?
- Where is the nearest open shelter?
- How do I leave if my phone has no signal?
- What do I do if I can't read the alert?

The bottleneck is not forecasting. It is the human and logistical chain between a warning and a person reaching safety.

Five Paths Forward

Each path targets a distinct documented failure from July 2024. These are options to evaluate – not a ranked recommendation.

Last-Mile Relay

When towers go down, relay via **SMS chains, community radio, or mesh networks** (Bluetooth / LoRa) so the alert reaches people even without cellular coverage

Hyperlocal Risk

Street-by-street or village-by-village flood forecasts – not just district-level river readings that average out local danger

Voice in Local Languages

IVR calls in **Assamese, Bodo, Khasi, Garo, Mizo** – reaching people without smartphones or with low literacy who are missed by app-based alerts

Five Paths Forward (continued)

1

Two-Way Reporting

Village heads and trained volunteers report ground conditions – cracking hillsides, rising banks, blocked roads – **back into the system** to close the information loop

2

What-To-Do-Next Layer

Real-time rerouting around blocked roads **plus shelter-capacity tracking** – so an alert tells you not just that danger is coming, but exactly where to go

Why This Matters Now

The Gap Is Real and Recent

Upper Assam July 2024 is not a historical case study – it is a live example of the same failure pattern repeating in a system we assumed was working

Tech Alone Is Not the Fix

Experts are consistent: the solution pairs **forecasting with community participation and local knowledge** – not just better hardware or wider app coverage

Each Path Maps to a Documented Failure

Every solution path on the previous slides addresses a specific, traceable breakdown from July – not a hypothetical gap we're guessing at

What We Need to Decide

Which Gap First?

Do we start with last-mile connectivity, hyperlocal forecasting, language access, two-way reporting, or the what-to-do-next layer?

Who Do We Partner With?

ASDMA? NDMA? Local NGOs already embedded in flood-affected districts? Existing community radio networks?

Pilot Scope and Timeline?

Which district or corridor do we test in first? What does a credible pilot look like before the next monsoon season?

Next Steps

Select Path

Choose one solution and owner

Map Integration

Identify stakeholders and systems

Define Pilot Metrics

Set success criteria before monsoon

The goal leaving this meeting: one path chosen, one owner per step, and a clear timeline to pilot before the 2025 monsoon window opens.